AFWS: Angle-Free Weakly Supervised Rotating Object Detection for Remote Sensing Images

Junyan Lu, Qinglei Hu, *Senior Member, IEEE*, Ruifei Zhu, Yali Wei, and Tie Li

***Abstract*— Horizontal annotation-based weakly supervised rotating object detection is a research field that has just been explored. This concept is expected to have a transformative effect on the advancement of data-driven rotating object detection methods. Existing pioneering researches directly regress the rotating rectangle based on angle description, which mainly face two limitations under the weakly supervised framework: 1) the regression form of weakly supervised learning is redundant relative to the training objective, thereby increasing the difficulty of model training and 2) the training objective of self-supervised (SS) learning does not have a close logical relationship with the test metrics, which may result in loss of accuracy. Addressing the above issues, this article proposes an angle-free weakly supervised rotating object detection framework, whose salient points mainly include the following: 1) by improving an angle-free rotating object representation, the decoupling between horizontal and rotating regression parameters in describing rotating objects is achieved; 2) a weakly supervised learning pipeline that is completely equivalent to the common horizontal object detection is designed to effectively relieve the difficulty of model training; and 3) a geometrically intuitive SS learning loss function is introduced to bridge the gap between the training objective and testing metrics. Experimental results on multiple large-scale remote sensing datasets confirm that the accuracy of this method is superior to the state-of-the-art (SOTA) weakly supervised rotating object detection methods, and is competitive even with SOTA fully supervised-related works.**

***Index Terms*— Angle-free, circular intersection over union (IoU), rotating object detection, self-supervised (SS), weakly supervised.**

I. Introduction

ROTATING object detection plays a key role in the interpretation of complex scenes such as aerial remote sensing image, scene text, and industrial inspection, and has been extensively studied [1], [2], [3], [4]. Since the labeling of rotating bounding box (bbox) is quite time-consuming and labor-intensive, it has become a major pain point that restricts the progress of data-driven methods. Considering that existing object detection datasets are mainly annotated with horizontal bbox (HBox), how to directly perform weakly supervised learning based on the HBox to obtain the fine-grained rotating bbox (RBox) of the object has indeed become a meaningful exploration. In the context of this demand, an ingenious and inspiring work was born: H2RBox [5]. H2RBox is the first HBox-supervised rotating object detection method. It pioneers a dual-branch supervised learning framework with model parameter sharing, in which the weakly supervised (WS) branch learns only by relying on the HBox annotation to give a coarse RBox prediction, and the self-supervised (SS) branch uses the prediction consistency between randomly rotated views of the object as constraints to improve the performance of the WS branch. With the continuous mutual promotion and iterative optimization of the WS and SS branches, H2RBox is expected to reach a stable and reliable state. On this basis, H2RBox-v2 [6] came into being. In addition to the rotating view (RV), it adds a flipping view to learn the symmetry of the object. It exploits the prediction consistency constraints between the original, rotating, and flipping views for SS learning, thus improving the accuracy and robustness of the model. The H2RBox series provides new ideas and approaches for HBox-based WS rotating object detection. However, as pioneering works, they inevitably expose some problems that deserve further exploration. This article mainly summarizes them into two aspects.

First, the WS branch of H2RBox involves angle regression, but in fact the angle is redundant for the training objective of this branch. Specifically, the regression subnetwork of the WS branch predicts HBox and angle parameters, respectively. At the same time, an additional decoder has to be trained to integrate the separate HBox and angle representations into the RBox representation with intuitive geometric meaning. However, in the end, H2RBox selects the circumscribed horizontal rectangle of the decoded RBox to calculate the intersection over union (IoU) loss with the HBox ground truth (GT) as the training objective, as shown in Fig. 1 (top). In other words, H2RBox first predicts the angle, then trains a decoder to integrate the angle, and finally discards the angle. This seems to be two useless efforts going back and forth. This article believes that both the angle prediction and the decoder are redundant in the WS branch. The addition of these two items increases the training difficulty of the model to a considerable

Received 6 November 2023; revised 6 June 2024 and 28 July 2024; accepted 18 October 2024. Date of publication 23 October 2024; date of current version 4 November 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62203036, in part by Zhejiang Provincial Natural Science Foundation under Grant LQ23F010025, in part by Shanghai Aerospace Science and Technology Innovation Foundation under Grant SAST2022-001, and in part by the Tianmushan Laboratory Research Project under Grant TK-2023-B-010 and Grant TK-2023-C-020. *(Corresponding author: Qinglei Hu.)*

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Digital Object Identifier 10.1109/TGRS.2024.3485590

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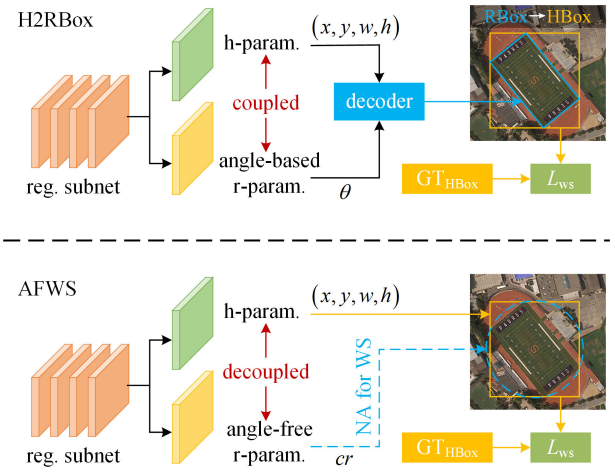


Fig. 1. Calculation of the weakly supervised loss L_{ws} in the angle-based regression and angle-free regression. (Top) H2RBox first predicts horizontal parameters (h-param.) (x, y, w, h) and angle-based rotating parameter (r-param.) θ separately, then integrates the two into an RBox through a decoder, and finally takes the HBox surrounding the RBox to calculate the L_{ws} with the HBox GT. (Bottom) AFWS proposed in this article directly uses the initially predicted horizontal parameters to calculate the L_{ws} with the HBox GT, while the angle-free rotating parameter prediction cr (representing the radius of the concentric circle) is not applicable for the WS branch, thus achieving the decoupling of the horizontal and rotating parameters.

extent, thereby affecting the detection accuracy. H2RBox-v2 is a similar story, whose WS branch also trains the angle parameter and decoder (which is called BoxCoder in H2RBox-v2) that are not directly involved in the loss calculation. In addition, the model training of H2RBox-v2 is more difficult than that of H2RBox. On the one hand, it adds a flipping view, causing the model optimization to be affected by the mutual influence of the three views. On the other hand, the regressors of its WS and SS branches are heterogeneous and cannot share parameters. Therefore, simplifying training parameters is more urgent for H2RBox-v2.

Second, H2RBox and H2RBox-v2 approximate or replace the RBox IoU loss by more than one step, leaving a gap between the training objective and testing metrics. In particular, H2RBox splits the RBox IoU loss used to supervise the SS branch into the $L1$ loss of (x, y) and the regression loss of (w, h, θ), and the latter is split into the HBox IoU loss of (w, h) and the regression loss of θ. Such two splits are equivalent to two approximations. This article believes that the authors do this to avoid the problem that RBox IoU is not differentiable as a loss function [10]. In fact, the loss function of the SS branch of H2RBox is also strictly non-differentiable, but the selection function and the absolute value function only have a small number of finite points that are not differentiable and belong to the first type of discontinuity points. Therefore, in engineering, they can be ignored after simple processing to achieve backpropagation. However, this article believes that this splitting manner loses the original geometric interpretability of the RBox IoU, making it more or less not logically relevant to the RBox IoU loss. H2RBox-v2 projects two rectangles in the WS branch to a parallel direction to calculate the HBox IoU loss instead of the RBox IoU loss (once approximation), but the projected rectangle is not the prediction itself but its circumscribed rectangle (twice approximation). And similarly, this article believes that the introduced parallel projection IoU loss and the RBox IoU loss do not have a closely intuitive geometric logic relationship.

In summary, this article rethinks the rationality of angle-based regression for rotating object detection, especially under the WS framework. In fact, the above two aspects can be attributed to one fundamental reason: the horizontal parameter and the angle are strongly coupled in the angle-based RBox representation. Once the two are separated, their geometric meanings are completely changed. In particular, as shown in Fig. 1 (top), H2RBox cannot directly use the initial regressed horizontal parameters to calculate the WS loss with the HBox GT, because the geometric meaning of the initial HBox is ambiguous. It has the geometric meaning of describing the object shape only after being combined with the angle and decoded. Therefore, the angle parameter and the decoder have to be involved in the training whose objective actually does not involve the angle, thus creating a burden on the model. On the contrary, in order to avoid the non-differentiability of the RBox IoU loss, H2RBox ignores the coupling relationship between the horizontal and angle parameters, and forcibly separates the two to calculate the SS loss, resulting in the inexplicability of the geometric meaning. In other words, the WS loss calculation and SS loss calculation in H2RBox seem to be contradictory in their understanding and processing of the coupling between the horizontal and rotating parameters.

Addressing above issues, this article proposes a novel angle-free weakly supervised framework AFWS for rotating object detection. AFWS goes directly to the root cause of the problem that angle-based regression makes the WS framework suffer, namely the strong coupling of the horizontal and rotating parameters, and adopts an angle-free regression form (which will be elaborated in Section III-A) to completely decouple the two. The angle-free regression can smoothly adapt to the training of both WS and SS branches, making the loss calculations of the two no longer contradictory. In particular, as shown in Fig. 1 (bottom), the initial predicted horizontal parameters of the regression subnet of AFWS have explicit and intuitive geometric sense and can be directly used to calculate the WS loss. While the angle-free rotating parameter cr (representing the radius of the concentric circle) is not applicable for the WS branch and does not play any role. By this way, the WS branch of AFWS can only focus on the training of horizontal parameter and fully enjoy the ready-made horizontal object detection pipelines, which effectively eliminates model redundancy and significantly reduces the difficulty of training. In addition, the introduced angle-free rotating object representation converts the angular description of RBox into a circular fashion. It is important to know that the essence of the H2RBox series to achieve weak supervision lies in the prediction consistency constraint between randomly rotated views, which also leads to the inevitable approximation or replacement of the RBox IoU loss. However, unlike rectangles, circles are rotationally invariant. On this basis, AFWS can design an SS loss function with more intuitive geometric meaning and closer logical correlation with the RBox IoU loss, effectively bridging the gap between the training objective and the testing metrics. As a result, AFWS gives more satisfactory performance in remote sensing rotating object detection than the state-of-the-art (SOTA) HBox-supervised methods, and the qualitative improvements can be seen in Fig. 2.

In summary, there are two main contributions of this article.

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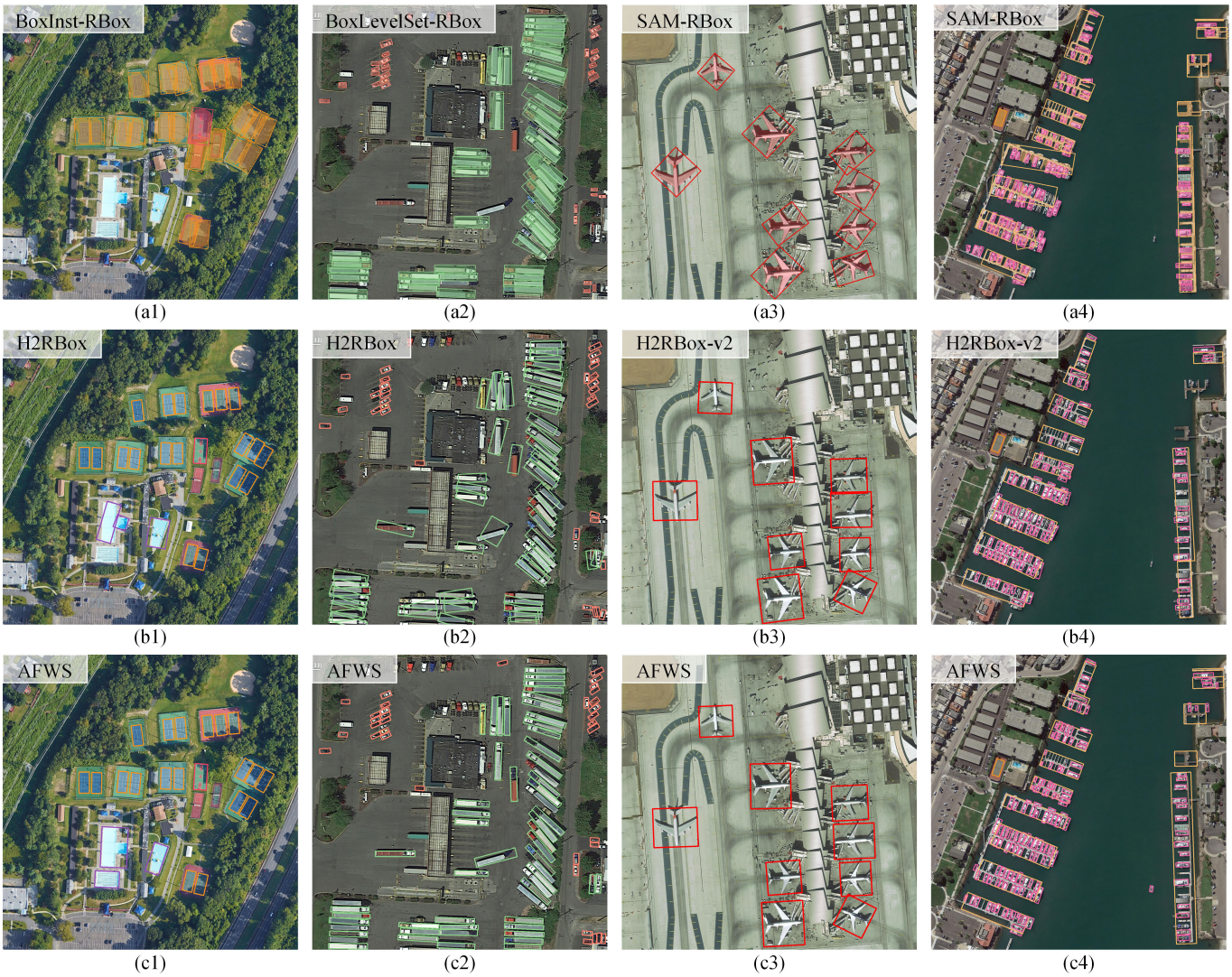


Fig. 2. Visual comparison of the effects of HBox-supervised rotating object detection methods. (a1)–(a4) Post-process the instance segmentation results to obtain the rotating bbox, and the methods include BoxInst [7], BoxLevelSet [8], and SAM [9]. (b1)–(b4) Rotating object detection results of H2RBox [5] and H2RBox-v2 [6]. (c1)–(c4) Rotating object detection results of AFWS proposed in this article.

1) A new idea of an angle-free regression form combined with the WS framework is introduced, which achieves the decoupling of the horizontal and rotating parameters, allowing the two to play a leading role in the training of the WS branch and the SS branch, respectively, thereby effectively simplifying the model training process and greatly easing the difficulty of model training.
2) An SS loss function with a more intuitive geometric meaning and a closer logical connection with the RBox IoU loss is designed, which innovatively utilizes the rotation invariance of the circle to cope with the RV of the SS branch, thus effectively bridging the gap between the training objective and the testing metrics and improving model accuracy.

The rest of this article is organized as follows. Section II introduces the related works. The proposed methodology is elaborated in Section III. Experimental results are reported and discussed in Section IV. Finally, Section V draws the conclusions.

II. Related Works

A. Studies on the Limitations Faced by Angle-Based RBox-Supervised Methods

A large number of studies have shown that directly regressing the angle of the RBox will mainly face two limitations. One is regression inconsistency and loss discontinuity caused by angle periodicity, and the other is the surge in regression loss caused by the boundary problem. Existing researchers have made constructive attempts to address such issues from multiple perspectives, mainly including the following:

1) Directly modulate the loss function to eliminate or alleviate the surge of regression loss. For example, SCRDet [11] introduces an IoU constant factor for the smooth $L1$ loss function to make the loss increment in the boundary case approximately 0; Qian et al. [12] design boundary constraints for $L1$ loss to eliminate jump discontinuities.
2) Encode angles to achieve loss continuity for boundary cases. In particular, CSL [13] and DCL [14] convert

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angle regression into discretized dense classification to ensure that the predictions do not exceed the definition domain; PSC [15] maps angles to cosines of different frequencies and phase offsets to ensure loss continuity.

3) Abstract the RBox into a probability distribution and use distribution distance instead of regression loss. For example, GWD [16] approximates the RBox as a 2-D Gaussian distribution and uses the Wasserstein distance as the loss; KLD [17] replaces Wasserstein distance with KL divergence based on GWD; KFIoU [18] simulates the calculation process of IoU through Gaussian distribution to achieve fully differentiable backpropagation.
4) Design a new RBox representation to avoid angle regression and fundamentally get rid of related limitations. In particular, Gliding Vertex [19] represents the RBox by predicting the offsets of the four vertices of the horizontal box; SAR [20] utilizes the geometric relationship between the concentric horizontal rectangle, circle, and rotating rectangle to design regression parameters.

B. Existing or Equivalent HBox-Supervised Rotating Object Detection Methods

1) The H2RBox series utilizes prediction consistency constraints between multiple views of the object to implement WS learning, and its overall framework is a fashion of object detection. In particular, H2RBox directly adds angle regression to the head of FCOS, and decodes the horizontal regression parameters combined with the angle regression parameter to obtain the RBox. In the WS branch, the IoU loss is calculated between the circumscribed HBox of the RBox prediction and the horizontal label. In the SS branch, the approximate RBox IoU loss is calculated between the RBox predictions of the original and RVs. On the basis of H2RBox, H2RBox-v2 adds a flipping view. Its SS branch learns the symmetry of the object based on the interrelationship between the original, rotating, and flipping views to give angle predictions, and its WS branch is roughly the same as H2RBox except that a direction-projection IoU loss is introduced.
2) The HBox-based SS instance segmentation has been relatively extensively studied. Taking the minimum circumscribed rectangle of the instance segmentation mask as the RBox can formally realize rotating object detection. Although this complicates a simple problem, it is an equivalent solution. For example, BoxInst [7] uses the discrepancy between the projections of the mask and the bbox, and the color affinity between adjacent pixels to design a loss function that does not rely on mask annotation. BoxLevelSet [8] treats the mask of each instance as the level set function of its corresponding object, which formulates image segmentation as a continuous energy minimization problem, and utilizes the input image and deep features to evolve the level set so that the level set for each instance is iteratively optimized within the bbox of the instance. SAM [9] expands the prompts in the diffusion model from text to segmentation conditions such as bboxes and foreground/background points, and returns effective segmentation masks under the given prompts.

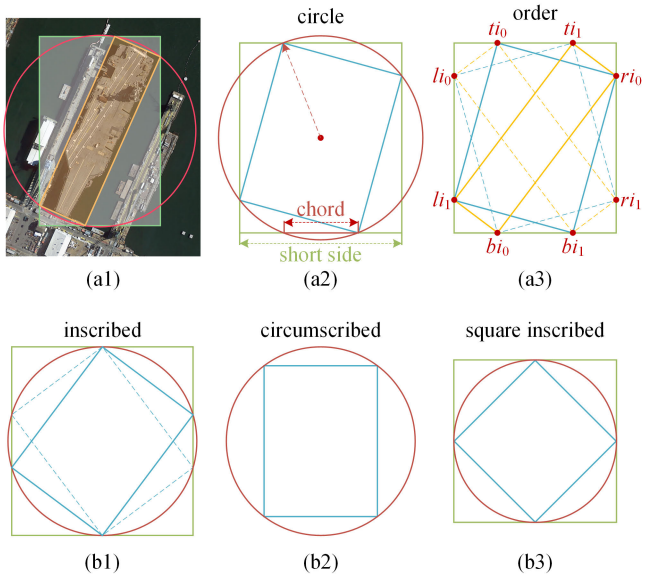


Fig. 3. Lite-CCH representation. (a1) Relationship between the HBox of the object, the RBox of the object, and the concentric circle; (a2) Meaning of the circle radius parameter; (a3) Corresponding relationship between the intersection order and the RBox; (b1)–(b3) three special cases where the circle inscribes the short side of the HBox, the circle circumscribes the HBox, and the circle is inscribed in a horizontal square, respectively.

Existing HBox-supervised methods have achieved remarkable results. However, on the one hand, for object detection-based methods (H2RBox series), their WS process is complex and redundant, and the SS training objective is not designed to be straightforward, as discussed in detail in the introduction. On the other hand, for instance segmentation-based methods, the instance segmentation pipeline is more complex and difficult than object detection. However, the fine-grained nature of its predictions is not actually needed but inevitably relies on more expensive time and computational costs. Therefore, the instance-to-RBox paradigm is not a wise and efficient choice.

III. Proposed Method

A. Lite-CCH Rotating Object Representation

The early work of this article proposed a rotating object representation that utilizes circle cutting horizontal rectangle (CCH) [20]. This article simplifies and improves CCH and introduces lite-CCH representation, as shown in Fig. 3. Specifically, on the premise that, both the horizontal and rotating bboxes of the object are rectangles, an RBox inscribed in the HBox annotation can be used as the rotating annotation of the object. In other words, when an RBox is used as the rotating annotation of the object, the circumscribed HBox of the RBox is appropriate as the horizontal annotation of the object. At the same time, the RBox must be inscribed in a concentric circle of the HBox (this conclusion is obvious and the proof will not be given here), as shown in Fig. 3(a1). Therefore, based on the HBox of the object, lite-CCH leverages the concentric circle cutting the HBox and the order of their intersections to represent the rotating bbox of the object.

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1) Concentric Circle: Given a HBox, by controlling the radius of its concentric circle, any RBox inscribed in the HBox can be obtained. The concentric circle inscribing the short side of the HBox corresponds to the minimum radius, and the concentric circle circumscribing the HBox corresponds to the maximum radius. Therefore, lite-CCH designs a circle parameter cr to represent the radius: $\mathrm{cr} = \mathrm{cd}/\mathrm{sl}$, where cd is the chord length obtained by the concentric circle cutting the short side of the HBox, and sl is the short side length of the HBox, as shown in Fig. 3(a2). This design can make $\mathrm{cr} \in [0, 1]$, that is, when $\mathrm{cr} = 0$, the circle inscribes the short side of HBox, and when $\mathrm{cr} = 1$, the circle circumscribes the HBox, so cr can be easily predicted through a Sigmoid layer.

2) Intersection Order: Generally, the circle and the HBox have eight intersections, two on each side of the HBox. Through these eight intersections, four RBoxes can be obtained, as shown in Fig. 3(a3). Lite-CCH introduces an order parameter **or** to uniquely determine an RBox. Specifically, the intersections of the top and bottom sides of the HBox are denoted as ti and bi, respectively; the smaller abscissa is marked as 0, and the larger one is marked as 1. The intersections of the left and right sides of the HBox are denoted as li and ri, respectively; the smaller ordinate is marked as 0 and the larger one is marked as 1. The image coordinate system used in this article is defined as follows: the upper left corner of the image is the origin, the horizontal right is the positive direction of the x-axis, and the vertical downward is the positive direction of the y-axis. **or** is defined as a 4-D one-hot vector, corresponding to the four RBoxes in general, where $\mathbf{or} = (1, 0, 0, 0)$ corresponds to the RBox $(\mathrm{ti}_1, \mathrm{li}_1, \mathrm{bi}_0, \mathrm{ri}_0)$ (yellow solid line), $\mathbf{or} = (0, 1, 0, 0)$ corresponds to the RBox $(\mathrm{ti}_0, \mathrm{li}_0, \mathrm{bi}_1, \mathrm{ri}_1)$ (yellow dotted line), $\mathbf{or} = (0, 0, 1, 0)$ corresponds to the RBox $(\mathrm{ti}_0, \mathrm{li}_1, \mathrm{bi}_1, \mathrm{ri}_0)$ (blue solid line), and $\mathbf{or} = (0, 0, 0, 1)$ corresponds to the RBox $(\mathrm{ti}_1, \mathrm{li}_0, \mathrm{bi}_0, \mathrm{ri}_1)$ (blue dotted line). **or** is defined in this way so that cross entropy can be used conveniently and reasonably to calculate its loss in supervised learning. Although this point is not covered in this article, lite-CCH as a representation itself should take this into account.

3) Special Cases: The special cases of lite-CCH are shown in Fig. 3(b1)–(b3). Among them, Fig. 3(b1) is the case that the circle inscribes the short sides of the HBox; at this time, $\mathrm{cr} = 0$, and **or** corresponds to two GTs. Fig. 3(b2) is the case that the circle circumscribes the HBox; at this time, $\mathrm{cr} = 1$, and **or** corresponds to all four GTs. Fig. 3(b3) is the case that the circle inscribes a horizontal square; at this time, $\mathrm{cr} = 0$, and **or** corresponds to all four GTs. Note that Fig. 3 is a drawing illustration using the width of the rectangle as the short side. If the height of the rectangle is the short side, the above principles are exactly the same and will not be repeated here.

4) Lite-CCH Versus CCH: The CCH proposed in the previous work of this article needs to predict two classification parameters, which, respectively, characterize the direction and width of RBox. In fact, there is a small flaw in the CCH-based representation, that is, the GT corresponding to the classification prediction is not unique near the special cases mentioned above (when cr is close to 0 or 1). However, multiobjective optimization usually limits the performance of the classifier, and the training of two classifiers faces this problem under the CCH representation. Therefore, the lite-CCH integrates the two classification parameters in CCH into one. This can make the model more streamlined on the one hand, and effectively alleviate the ambiguity around special cases on the other hand, which is indeed beneficial to improving model accuracy.

B. Angle-Free Weakly Supervised Rotating Object Detection Framework

1) Overall Architecture: Based on the lite-CCH representation and inspired by H2RBox, this article proposes an angle-free weakly supervised rotating object detection framework AFWS, as shown in Fig. 4. AFWS adopts a Siamese-like [21] architecture and consists of two parallel branches, the WS branch and the SS branch, which are conceptually similar to H2RBox. The two branches do parallel inferences and share the same set of network parameters, including the backbone, neck, and head. The AFWS network generally employs a common horizontal object detection network structure. As demonstrated in Fig. 4, if the rotating parameters "circle" and "order" in the gray dashed box are not considered, it is a vanilla horizontal object detection model. The AFWS head contains two subnets, a classification subnet and a regression subnet. The classification subnet predicts two sets of parameters, one is a C-dimensional vector **p** representing the object category (C is the number of categories), and the other is a 4-D one-hot vector **or** representing the RBox order in the lite-CCH representation. The regression subnet predicts two sets of parameters, one is a 4-D vector **hb** indicating the HBox, and the other is the circle radius parameter cr in the lite-CCH representation. The training of AFWS is completed jointly by the WS and SS branches. The input of the two branches includes the different views of the same image and is predicted separately. The WS branch is responsible for training **hb** and **p**, and the SS branch is responsible for training **hb**, cr, and **or**, where **hb** participates in training of both branches. The WS branch learns only with horizontal annotations, and the SS branch learns with predictions from dual views. Based on the AFWS architecture, in theory, any horizontal object detection benchmark can be used to achieve rotating object detection, which is universal and plug-and-play. It should be noted that the AFWS baseline proposed in this article takes advantage of the FCOS benchmark, and the FCOS head contains a center-ness (cn) parameter prediction. However, Fig. 4 is intended to illustrate the generality of the AFWS framework and emphasize the lite-CCH-based rotating object detection principle. For the sake of clarity and simplicity of composition, cn is not reflected. In fact, the cn prediction is located in the regression subnet of the AFWS baseline head, and its meaning is the same as in FCOS.

2) Weakly Supervised Branch: The input of the WS branch is the original image which is called the original view (OV) in this article, and the output is called the original view prediction (OVP). In the training phase, the WS branch only uses the predicted **p** and **hb** parameters for supervised learning, and the required GT is solely the horizontal annotation labels, that is, the training process of the WS branch is equivalent to an ordinary horizontal object detection network. In the test phase, although cr and **or** are ignored in the training phase, they

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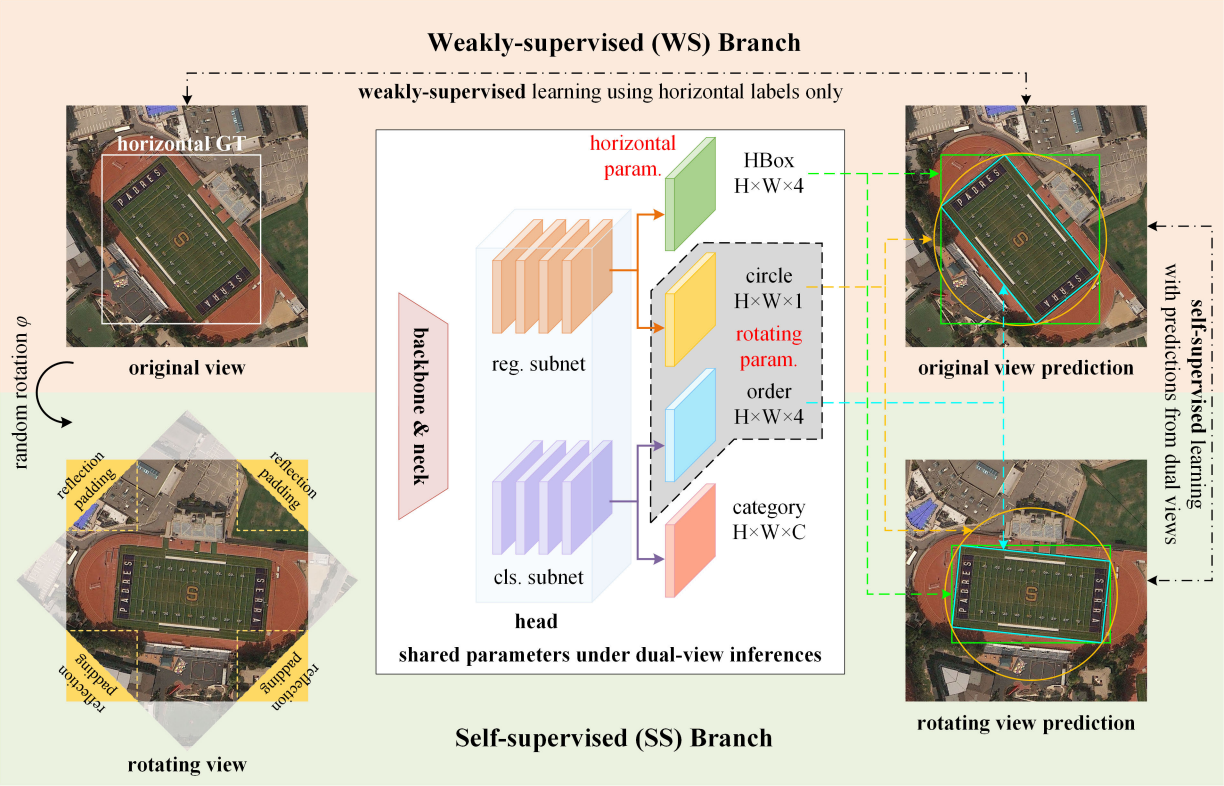


Fig. 4. AFWS framework. Top and bottom parallel parts of the figure are the WS and SS branches of AFWS, respectively, both of which share the same set of network parameters, including the backbone, neck, and head. The head contains two subnets: a classification subnet and a regression subnet, where the classification subnet predicts the order parameter in the lite-CHH representation and the category parameter, and the regression subnet predicts the circle parameter in the lite-CHH representation and the HBox parameter, while the order and circle make up the rotating parameters. H and W represent the height and width of the feature map, respectively; C represents the number of object categories, and the definition of "reflection padding" is the same as in H2RBox. In the training phase, the input of the WS branch is the original view of an image, and this branch only uses the horizontal GT training the category and HBox parameters. The input of the SS branch is a randomly rotated view of the same image, which uses the prediction of the WS branch as GT to train the order and circle parameters. In the testing phase, the output of the WS branch is the prediction of AFWS.

play a role in the test phase. The four predicted parameters perform their duties for giving the rotating bbox and category of the object. The OVP is also the final prediction of AFWS in the test phase. Note that the AFWS baseline trains the cn parameter in the WS branch.

3) SS Branch: The input of the SS branch is the original image after random rotation and corresponding processing, which is called the RV in this article, and the output is called the RV prediction (RVP). The OV randomly rotates φ around the center of image, and then it is cropped according to the size range of the original image, and the blank area is filled with reflection padding to obtain the RV. The rotation transformation relation between OV and RV is as shown in the following equation:

$$(x', y') = (x - x_0, y - y_0)\mathbf{R}^\top + (x_0, y_0)$$
$$\mathbf{R} = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix} \tag{1}$$

where (x, y) represents any location of the image or feature map in the WS branch, (x', y') represents the corresponding location of (x, y) of the image or feature map in the SS branch, and (x_0, y_0) is the image center of the OV. φ is randomly generated between $[\pi/4, 3\pi/4]$,[1] and reflection padding is the same as in H2RBox. In the training phase, the SS branch takes the consistency of the OVP and RVP as the criterion for learning. Specifically, the role of SS branch is to train the network so that it has the ability to give the same prediction when facing different views of the same object. The "same prediction" means that there is an explicit rotation transformation relation between the geometric and position parameter predictions of the object under different views. Based on the lite-CCH representation, the dual-view prediction consistency loss of the SS branch is shown in Fig. 5. For the three subgraphs in Fig. 5, the left, right, and middle are the OVP, RVP, and original view prediction transformation (OVPT), respectively. Taking the "football field" object in each subgraph as an example, the RBox, HBox, and circle parameters of the object are represented by the corresponding **rb**, **hb**, and **cr**, respectively. Taking **rb** as an example, the RBox parameters in OVP, RVP, and OVPT are represented as $\mathbf{rb}^{\mathrm{WS}}$, $\mathbf{rb}^{\mathrm{SS}}$, and $\mathbf{rb}^{\mathrm{SS*}}$, respectively. Pay attention to the distinction between **cr** here and cr in Section III-B1, where **cr** is a vector containing the center position and radius parameters, while cr is a scalar representing the radius of the circle. In particular, the $\mathbf{rb}^{\mathrm{SS*}}$ in OVPT is obtained by the rotation transformation of $\mathbf{rb}^{\mathrm{WS}}$ through (1), and then the corresponding $\mathbf{hb}^{\mathrm{SS*}}$ and $\mathbf{cr}^{\mathrm{SS*}}$ can be deduced from $\mathbf{rb}^{\mathrm{SS*}}$. The above rotation transformation and deduction only involve basic analytic geometry calculations and therefore will not be elaborated. $\mathbf{rb}^{\mathrm{SS*}}$, $\mathbf{hb}^{\mathrm{SS*}}$, and $\mathbf{cr}^{\mathrm{SS*}}$ can be considered as the GTs of $\mathbf{rb}^{\mathrm{SS}}$, $\mathbf{hb}^{\mathrm{SS}}$, and $\mathbf{cr}^{\mathrm{SS}}$, respectively.

[1]It is pointed out in H2RBox-v2 that the range of the random rotation angle will affect the stability of the SS branch training, and it performs well when the angle ranges in $[\pi/4, 3\pi/4]$. This article refers to the conclusion.

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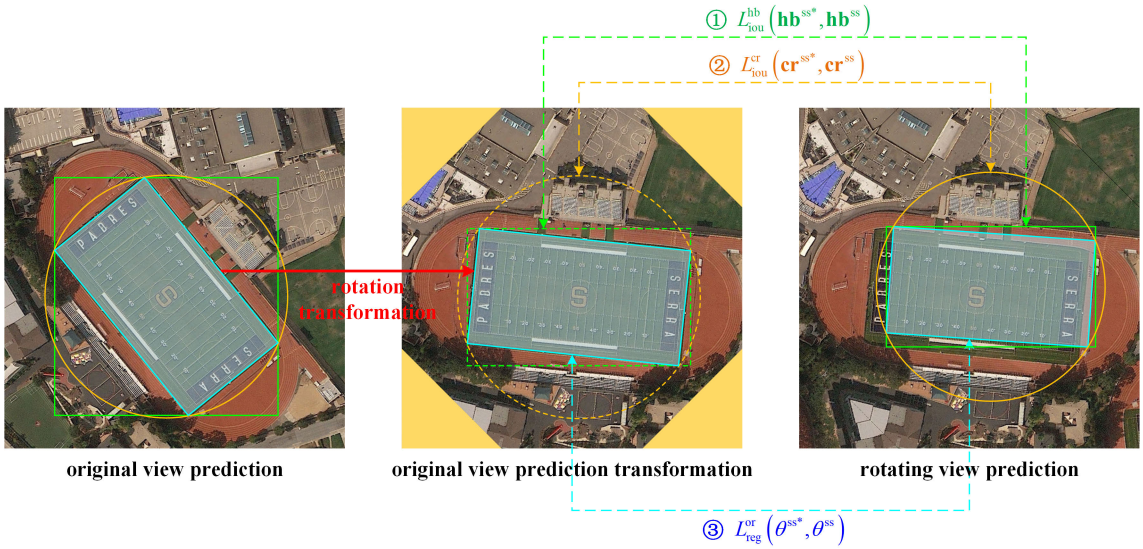


Fig. 5. Loss in the SS branch. The subgraphs left, middle, and right indicate OVP, OVPT, and RVP, respectively. The RBox (cyan solid line) in OVPT is the rotation transformation of the RBox in the OVP, and based on this, the HBox (green dashed line) and circle (yellow dashed line) in the OVPT can be deducted. The OVPT is regarded as the GT of RVP to train the SS branch, and its loss consists of three parts: ① IoU loss of the HBoxes between the OVPT and RVP (L_{iou}^{hb}), ② IoU loss of the circles between the OVPT and RVP (L_{iou}^{cr}), and ③ rotation angle loss of the RBoxes between the OVPT and RVP (L_{reg}^{or}).

That is, the GT of RVP is not prepared in advance, but the OVPT generated based on OVP during training. Ideally, $\mathbf{hb}^{ss*}$ and $\mathbf{hb}^{ss}$, $\mathbf{cr}^{ss*}$ and $\mathbf{cr}^{ss}$, and $\mathbf{rb}^{ss*}$ and $\mathbf{rb}^{ss}$ should all coincide, that is, AFWS achieves complete consistency in predictions from different views. Therefore, the SS branch integrates the $L_{iou}^{hb}(\mathbf{hb}^{ss*},\mathbf{hb}^{ss})$, $L_{iou}^{cr}(\mathbf{cr}^{ss*},\mathbf{cr}^{ss})$, and $L_{reg}^{or}(\theta^{ss*},\theta^{ss})$ for learning. In particular, L_{iou}^{hb} is the conventional IoU loss [22] of HBoxes $\mathbf{hb}^{ss*}$ and $\mathbf{hb}^{ss}$, which is partially responsible for training the **hb** parameter of AFWS. By analogy, L_{iou}^{hb} can be generalized to the circular IoU [23] loss L_{iou}^{cr} to represent the coincidence degree of circles $\mathbf{cr}^{ss*}$ and $\mathbf{cr}^{ss}$. The circular IOU is shown in Fig. 6, and L_{iou}^{cr} is responsible for training the cr parameter of AFWS. L_{reg}^{or} represents the regression loss between θ^{ss*} and θ^{ss}, where θ^{ss*} and θ^{ss} are the rotation angles of RBoxes $\mathbf{rb}^{ss*}$ and $\mathbf{rb}^{ss}$ under the angle-based representation. The angle-based representation is defined as $\mathbf{rb} = (x, y, w, h, \theta)$, where (x, y) is the geometric center of the RBox, the set of sides that form an acute angle with the positive direction of the x-axis is the width (w), and the other set of sides is the height (h); the angle is positive in the clockwise direction, so the angle domain is $(-\pi/2, 0]$. The above definition is the same as that in OpenCV, which is for the convenience of code implementation. If the **or** predictions in OVP and RVP are correct, then θ^{ss*} and θ^{ss} should be the same, so L_{reg}^{or} is responsible for training the **or** parameter of AFWS. For the detailed definition of the loss in the SS branch, please refer to Section III-B4. The SS branch does not involve the training of the category parameter **p**, because this article finds in the ablation experiments that adding the category loss to the SS branch for learning will hardly improve the overall classification accuracy of AFWS, that is, only training **p** in the WS branch can already achieve satisfactory results.

4) Training Objective: AFWS is a general architecture that takes advantage of horizontal object detection benchmarks to achieve rotating object detection. Therefore, its training objective also involves corresponding horizontal object detection loss functions, which may be different for various specific methods. This section elaborates the training objective for the FCOS-like AFWS, which is the baseline proposed in this article. In general, AFWS combines the losses of the WS and SS branches for joint learning, that is, the overall training objective L of AFWS is as follows:

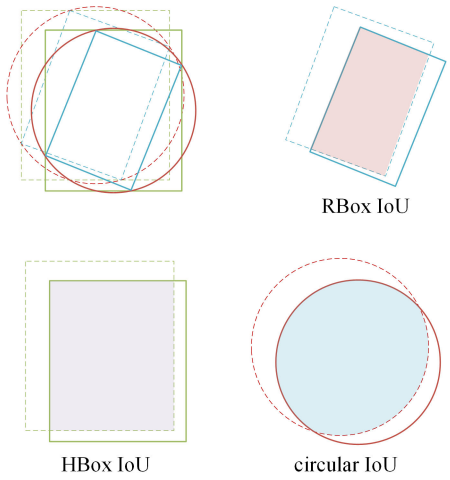


Fig. 6. Schematic of the RBox IoU, HBox IoU, and circular IoU.

$$L = L_{ws} + \lambda L_{ss} \tag{2}$$

where L_{ws} is the loss in the WS branch, L_{ss} is the loss in the SS branch, and λ is the balance weight, which is set to 0.25 by default. L_{ws} is defined as follows:

$$\begin{aligned} L_{ws} = & \frac{1}{N_{pos}^{ws}} \sum_{(x,y)} L_{cls}\left(\mathbf{p}_{(x,y)}^{*}, \mathbf{p}_{(x,y)}\right) \\ & + \frac{\beta_1}{N_{pos}^{ws}} \sum_{(x,y)} L_{reg}^{cn}\left(cn_{(x,y)}^{*}, cn_{(x,y)}\right) \\ & + \frac{\beta_2}{N_{pos}^{ws}} \sum_{(x,y)} \mathbb{1}_{\left\{\mathbf{p}_{(x,y)}^{*} \neq \mathbf{0}\right\}} L_{iou}^{hb}\left(\mathbf{hb}_{(x,y)}^{ws*}, \mathbf{hb}_{(x,y)}^{ws}\right) \end{aligned} \tag{3}$$

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where (x, y) denotes each location of the feature maps in the WS branch; $N_{\text{pos}}^{\text{ws}}$ is the number of positive samples in the WS branch, and the definition of positive sample in the WS branch is the same as in FCOS; L_{cls} is the focal loss [24], $\mathbf{p}^*$ and $\mathbf{p}$ are the category GT and the category prediction in WS branch, respectively; $L_{\text{iou}}^{\text{hb}}$ is the HBox IoU loss [22], $\mathbf{hb}^{\text{ws}*}$, and $\mathbf{hb}^{\text{ws}}$ are the horizontal bbox GT and the HBox prediction in the WS branch, respectively; $\mathbf{hb}$ adopts the HBox form in FCOS, that is, $\mathbf{hb} = (l, t, r, b)$; $L_{\text{reg}}^{\text{cn}}$ is the BCE loss, cn* and cn are the center-ness GT and the cn prediction in the WS branch, respectively, and the calculation of cn^* is the same as in FCOS; $\mathbb{1}_{\{\mathbf{p}^*_{(x,y)} \neq \mathbf{0}\}}$ is the indicator function, being 1 if $\mathbf{p}^*_{(x,y)} \neq \mathbf{0}$ and 0 otherwise; β_1 and β_2 are the balanced weights, which are all set to 1 by default; L_{ws} is equivalent to a general common horizontal object detection loss function, and cr and **or** parameters are not involved in the WS branch training. L_{ss} is defined as follows:

$$\begin{aligned} L_{\text{ss}} = & \frac{1}{N_{\text{pos}}^{\text{ss}}} \sum_{(x',y')} \mathbb{1}_{\left\{\mathbf{p}^*_{(x',y')} \neq \mathbf{0}\right\}} L_{\text{iou}}^{\text{hb}}\left(\mathbf{hb}^{\text{ss}*}_{(x',y')}, \mathbf{hb}^{\text{ss}}_{(x',y')}\right) \\ & + \frac{\gamma_1}{N_{\text{pos}}^{\text{ss}}} \sum_{(x',y')} \mathbb{1}_{\left\{\mathbf{p}^*_{(x',y')} \neq \mathbf{0}\right\}} L_{\text{iou}}^{\text{cr}}\left(\mathbf{cr}^{\text{ss}*}_{(x',y')}, \mathbf{cr}^{\text{ss}}_{(x',y')}\right) \\ & + \frac{\gamma_2}{N_{\text{pos}}^{\text{ss}}} \sum_{(x',y')} \mathbb{1}_{\left\{\mathbf{p}^*_{(x',y')} \neq \mathbf{0}\right\}} L_{\text{reg}}^{\text{or}}\left(\theta^{\text{ss}*}_{(x',y')}, \theta^{\text{ss}}_{(x',y')}\right) \end{aligned} \tag{4}$$

where (x', y') denotes each position of the feature maps in the SS branch, and the relation between (x', y') and (x, y) is explained by (1); $N_{\text{pos}}^{\text{ss}}$ is the number of positive samples in the SS branch, and please refer to Section IV-B for the definition of positive sample in the SS branch; $L_{\text{iou}}^{\text{hb}}$ is the HBox IoU loss, where $\mathbf{hb}^{\text{ss}}$ is the HBox prediction in the SS branch; $L_{\text{iou}}^{\text{cr}}$ is the circular IoU loss explained above, where $\mathbf{cr}^{\text{ss}}$ is the circle prediction in the SS branch; the form of $\mathbf{cr}$ is $\mathbf{cr} = (x_c, y_c, r_c)$, where (x_c, y_c) is the center coordinate of the circle (that is, the center of $\mathbf{hb}^{\text{ss}}$), and r_c is the circle radius (converted by cr); $\mathbf{hb}^{\text{ss}*}$ and $\mathbf{cr}^{\text{ss}*}$ are the intermediate GTs of $\mathbf{hb}^{\text{ss}}$ and $\mathbf{cr}^{\text{ss}}$ generated in the OVPT, respectively; $\theta^{\text{ss}*}$ and θ^{ss} are the rotation angles of RBoxes $\mathbf{rb}^{\text{ss}*}$ and $\mathbf{rb}^{\text{ss}}$, respectively; in fact, the RBox obtained based on lite-CCH representation is in the form of vertex coordinates: $\mathbf{rb} = (x_1, y_1, x_2, y_2, x_3, y_3, x_4, y_4)$, which is converted into the form of $\mathbf{rb} = (x, y, w, h, \theta)$ to calculate $L_{\text{reg}}^{\text{or}}$. The definition of $L_{\text{reg}}^{\text{or}}$ is as follows:

$$L_{\text{reg}}^{\text{or}}\left(\theta^{\text{ss}*}, \theta^{\text{ss}}\right) = \min\left\{\left|\sin\left(\theta^{\text{ss}*} - \theta^{\text{ss}}\right)\right|, \left|\cos\left(\theta^{\text{ss}*} - \theta^{\text{ss}}\right)\right|\right\}. \tag{5}$$

$L_{\text{reg}}^{\text{or}}$ is designed to deal with the boundary conditions under angle-based representation; specifically, the training objective expects $\theta^{\text{ss}*} - \theta^{\text{ss}} \approx \pm 0$, but when they are near the boundary of the angle definition domain $[-\pi/2, 0)$, a situation that may occur is that $\mathbf{rb}^{\text{ss}*}$ and $\mathbf{rb}^{\text{ss}}$ are close, but $\theta^{\text{ss}*}$ or θ^{ss} is forcibly changed because it is out of domain, resulting in $\theta^{\text{ss}*} - \theta^{\text{ss}} \approx \pm\pi/2$; (5) is able to give a reasonable loss in the above case. γ_1 and γ_2 are balance weights, and their default values are set to 2 and 1, respectively. The default value settings of hyperparameters λ, β, and γ mainly consider three aspects: 1) according to the principle of lite-CCH representation, RBox is obtained based on HBox, that is, this article believes that the horizontal parameter prediction effect has a greater impact on the final accuracy, so the WS branch takes a dominant position in training; 2) the β settings refer to FCOS; and 3) the SS branch is mainly responsible for the training of rotating parameters, and its effect mainly depends on the cr prediction, so $L_{\text{iou}}^{\text{cr}}$ is given greater weight. Subsequent ablation studies provide quantitative discussions on the settings of the above hyperparameters.

IV. Experiments

A. Datasets

1) DOTA: DOTA is a well-known large-scale aerial remote sensing image dataset. It has been developed and updated in three versions: DOTA-v1.0 [1], DOTA-v1.5 [25], and DOTA-v2.0 [26], which are released by the WHU in 2018, 2019, and 2021, respectively. In particular, DOTA-v1.0 collects 2806 satellite images from Google Earth, GF-2, and JL-1, and manually annotates 188 282 instances, covering 15 categories of objects: plane, baseball diamond, bridge, ground field track, small vehicle, large vehicle, ship, tennis court, basketball court, storage tank, soccer-ball field, roundabout, harbor, swimming pool, and helicopter. The annotation format is a four-vertex, eight-coordinate oriented bbox (OBB), and the corresponding horizontal bbox (HBB) is directly derived from OBB. The training, validation, and test sets are roughly divided into proportions of 1/2, 1/3, and 1/6. Based on DOTA-v1.0, DOTA-v1.5 adds a container crane category as well as annotations for tiny objects with a size less than 10 pixels, expanding the number of instances to 403 318. DOTA-v2.0 has further added a large number of aerial images provided by CycloMedia B.V., and adds two categories of objects (airport and helipad) increasing the number of images to 11 268 and the number of instances to 1 793 658. This article crops the original remote sensing images into 1024×1.024 patches with an overlap of 200 pixels during training and testing, and follows the procedures of dataset division, detection result merging, and model performance evaluation in the official DOTA benchmark.[2]

2) DIOR-R: DIOR-R [27] is a large-scale aerial remote sensing image dataset. Its predecessor DIOR [28] contains 23 463 images with size of 800×800 pixels, annotates 190 288 instances with HBB, covering 20 categories of objects; DIOR-R adds OBB annotations on this basis. DIOR-R focuses on the scale changes of objects within and between categories as well as imaging quality differences, resulting in a wide range of object diversity. This article follows the original training, validation, and test set division of the DIOR-R benchmark.

3) FAIR1M: FAIR1M-1.0 [29] is a large-scale dataset dedicated to refined object detection in aerial remote sensing images. It contains 16 488 training images and 8137 test images with size ranging from 1000×1000 to $10\,000 \times 10\,000$ pixels, space resolution ranging from 0.3 m to 0.8 m. FAIR1M-1.0 adopts OBB to annotate objects in five categories and 37 subcategories, where the subcategory is a detailed division of the major categories according to attributes such as type, function, size, etc. FAIR1M-1.0 has annotated more than one million instances. This article crops the original images into 1024×1.024 patches with an overlap of 200 pixels (the original image with a size less than 1024 is resized to a scale

[2]https://captain-whu.github.io/DOTA/code.html

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rate of 1.5), and follows the official FAIR1M benchmark[3] to merge and evaluate the detection results.

B. Implementation Details

1) AFWS Baseline: AFWS baseline model is built based on FCOS benchmark. Its backbone and neck adopt ResNet-50 [30] and FPN [31], respectively. The number of channels of the classification and regression subnets in the head is both 256. On the basis of the depiction in Fig. 4, AFWS baseline also predicts an additional cn parameter through a Sigmoid layer in the regression subnet. AFWS baseline is implemented based on MMRotate [32][4] under Python 3.7 and PyTorch 1.6.0. The provided H2RBox model[5] trained on DOTA-v1.0 is pruned accordingly and adopted as initial weight, and the newly added layers are initialized as in reference [24]. AdamW [33] is used as an optimizer with an initial learning rate of $5e^{-5}$. The learning rate is warmed up by 500 iterations and reduced tenfold at each decay step, and the weight decay is 0.05. AFWS baseline is trained on all datasets for 12 epochs or 36 epochs with a mini-batch size of 2. The model uses NVIDIA RTX 3090 GPU for training and testing.

2) AFWS Inference: The following points need to be noted during the inference of AFWS, especially its SS branch.

1) *Object Filtering of the RV:* The RV will crop out some areas in the OV. In this article, if the HBB GT of an object in the OV intersects with the cropped area, then the object will be filtered out in the RV. Objects contained in the reflection padding areas are ignored, while the reflection padding is used only to replace the zero padding without generating new objects.
2) *Positive Samples in the RV:* If the location (x, y) in the OV feature maps is a positive sample, then its corresponding location (x', y') in the RV is also a positive sample. However, if the object corresponding to location (x, y) is filtered out in the RV, then the location (x', y') is a negative sample.
3) *Endpoint Approximation of the cr Prediction:* During inference, due to the truncation error and data type conversion, negative value under square root may appear in analytic geometry calculations for some extreme cases; this article approximates the cr prediction close to the endpoint of its domain to avoid the situation: when $cr < 0.05$, let $cr = 0$; when $cr > 0.95$, let $cr = 1$.

3) Reproduction of Comparison Methods: This article aims to compare AFWS with the SOTA HBox-supervised and RBox-supervised rotating object detection methods under the same platform and settings, so as to objectively and comprehensively evaluate the performance of AFWS. In particular, the RBox-supervised comparison methods include RetinaNet [24] (rotated), FCOS [34] (rotated), RepPoints [35] (rotated), GWD [16], KLD [36], KFIoU [18], SASM [37], R^3Det [38], CFA [39], Oriented RepPoints [40], and S^2A-Net [41]. The HBox-supervised comparison methods include BoxInst [7], BoxLevelSet [8], SAM [9], H2RBox, and H2RBox-v2. All the above comparison methods are reproduced in this article. Among them, all the HBox-supervised methods are implemented based on MMRotate,[6] using ResNet-50 as the backbone, and their respective models trained on DOTA-v1.0 provided in MMRotate's Model Zoo are used as the initial weights. For the RBox-supervised methods, BoxInst[7] and BoxLevelSet[8] are implemented based on their respective original code, using ResNet-50 as the backbones, and the provided models trained on COCO are used as the initial weights. SAM[9] is implemented based on MMRotate-SAM, using ViT-B as the backbone, and the provided model trained on DOTA-v1.0 is used as the initial weight. For the above segmentation-based methods, this article obtains the minimum circumscribed rotation rectangle of the instance mask through post-processing, which can be regarded as the OBB of the object. Both H2RBox[10] and H2RBox-v2[11] are implemented based on MMRotate, using ResNet-50 as the backbones, and the models trained on DOTA-v1.0 provided by each of them are used as the initial weights. Other implementation details of the above comparison methods are the same as AFWS baseline. The BoxLevelSet model uses NVIDIA A100 GPU for training and testing due to excessive memory usage.

C. Results

1) DOTA-v1.0: The test results on DOTA-v1.0 of AFWS baseline and related works are reported in Table I. The "1×" and "3×" schedules in the table refer to training for 12 and 36 epochs, respectively. "MS" indicates that multiscale training and testing in MMRotate are adopted. AP_{50} performance is used to evaluate the accuracy of each model. The quantitative experimental results show that: *first*, the performance of AFWS basically exceeds the existing SOTA WS rotating object detection methods. In particular, compared with the object detection-based HBox-supervised methods, the AFWS baseline's 960-size+1×-schedule AP_{50} is 72.64%, which is 4.79% higher than H2RBox and 0.94% higher than H2RBox-v2, respectively; 1024-size+1×-schedule AP_{50} is 72.55%, which is 4.77% higher than H2RBox and 0.31% higher than H2RBox-v2, respectively; 1024-size+3×-schedule AP_{50} is 74.76%, 4.49% higher than H2RBox but 0.26% lower than H2RBox-v2; 1024-size+1×-schedule+MS AP_{50} is 78.13%, 3.37% higher than H2RBox and 0.25% higher than H2RBox-v2. Moreover, the AFWS baseline achieves an inference speed of 28.8 FPS (1024-size), which is not much different from H2RBox (29.4) and H2RBox-v2 (28.4) in efficiency. Compared with the instance segmentation-based methods, the AFWS baseline has achieved a huge lead in both accuracy and efficiency. Its AP_{50} is up to nearly 20% ahead of BoxInst-RBox (72.64% to 53.34%), and its frames per second (FPS) is up to more than 16× of SAM-ViT-B-RBox (30.7–1.9). In fact, applying BoxInst, BoxLevelSet, and SAM to rotating object detection complicates a simple problem. On the one hand, instance segmentation is inherently a more difficult task than object detection. On the other hand, converting instance segmentation masks into rotating bboxes involves time-consuming post-processing, and even

[3]https://gaofen-challenge.com/benchmark
[4]https://github.com/open-mmlab/mmrotate
[5]https://github.com/open-mmlab/mmrotate/tree/1.x/configs/h2rbox
[6]https://github.com/open-mmlab/mmrotate
[7]https://github.com/aim-uofa/AdelaiDet/tree/master/configs/BoxInst
[8]https://github.com/LiWentomng/boxlevelset
[9]https://github.com/open-mmlab/playground/tree/main/mmrotate_sam
[10]https://github.com/open-mmlab/mmrotate/tree/1.x/configs/h2rbox
[11]https://github.com/open-mmlab/mmrotate/tree/1.x/configs/h2rbox_v2

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TABLE I
TEST RESULTS ON THE DOTA-V1.0 DATASET

	Method	Sched.	MS	Size	FPS	AP_{50}
RBox-supervised	RepPoints [35]	1×		1024	24.9	68.03
	RetinaNet [24]	1×		1024	26.1	68.21
	GWD [16]	1×		1024	25.5	70.82
	KLD [36]	1×		1024	25.5	70.89
	KFIoU [18]	1×		1024	25.7	71.34
	SASM [37]	1×		1024	25.1	72.19
	R^3Det [38]	1×		1024	19.4	72.75
	CFA [39]	1×		1024	24.9	73.13
	Oriented RepPoints [40]	1×		1024	23.7	74.67
	S^2A-Net [41]	1×		1024	22.7	**75.49**
	FCOS [34]	1×		1024	**30.2**	71.93
	FCOS [34]	3×		1024	**30.2**	74.45
	FCOS [34]	1×	✓	1024	**30.2**	77.20
HBox-supervised	BoxInst-RBox [7][1]	1×		960	2.6	53.34
	BoxLevelSet-RBox [8][1]*	1×		960	5.1	56.72
	SAM-ViT-B-RBox [9][1]	1×		960	1.9	65.09
	H2RBox (FCOS-based) [5]	1×		960	**32.0**	67.85
	H2RBox (FCOS-based) [5]	1×		1024	29.4	67.78
	H2RBox (FCOS-based) [5]	3×		1024	29.4	70.27
	H2RBox (FCOS-based) [5]	1×	✓	1024	29.4	74.76
	H2RBox-v2 (FCOS-based) [6]	1×		960	29.1	71.70
	H2RBox-v2 (FCOS-based) [6]	1×		1024	28.4	72.24
	H2RBox-v2 (FCOS-based) [6]	3×		1024	28.4	75.02
	H2RBox-v2 (FCOS-based) [6]	1×	✓	1024	28.4	77.88
	AFWS (FCOS-based)	1×		960	30.7	72.64
	AFWS (FCOS-based)	1×		1024	28.8	72.55
	AFWS (FCOS-based)	3×		1024	28.8	74.76
	AFWS (FCOS-based)	1×	✓	1024	28.8	**78.13**

[1] "-RBox" means post-processing the segmentation mask to obtain its minimum circumscribed rotation rectangle.
* An NVIDIA A100 GPU is used due to the significant memory requirement of the model.

SAM requires additional pre-processing, that is, using FCOS to give rough horizontal detection results as its prompts. Therefore, the performances of the instance segmentation-based HBox-supervised methods seem to be unsatisfactory. The visualization results in Fig. 2 give a more intuitive comparison between the above methods. For the segmentation-based HBox-supervised methods [Fig. 2(a1)–(a4)], the uneven quality of instance segmentation may cause its bounding rectangle to not fit the object or even be outrageous. In addition, the disadvantages of instance segmentation in dealing with densely distributed small objects and huge differences in object scales are also clearly reflected in the figure. For H2RBox/H2RBox-v2 [Fig. 2(b1)–(b4)], Fig. 2(b2) shows that H2RBox faces obvious boundary problem in horizontal/vertical object detection, which is also one of the main flaws of the angle-based methods; Fig. 2(b4) shows that H2RBox-v2 effectively alleviates the boundary problem due to the use of a specialized angle encoder, while it still has room for improvement in densely distributed small object detection. The AFWS baseline results are depicted in Fig. 2(c1)–(c4), which shows that, thanks to the angle-free design, AFWS baseline effectively copes with the boundary problem and also performs well in detection of densely distributed small objects. *Second*, AFWS is able to achieve comparable rotating object detection accuracy with SOTA fully supervised methods through only WS learning. In particular, compared with the RBox-supervised methods, the 1×-schedule AP_{50} of AFWS baseline (72.55%) is lower than R^3Det (72.75%), CFA (73.13%), Oriented RepPoints (74.67%), and S^2A-Net (75.49%), while ahead of other methods. It is worth noting that the above superior methods have more or less performed some filtering or fusion processing on the feature layers. This article believes that these operations are essentially a kind of feature enhancement or attention mechanism, which can naturally enhance the effect of feature extraction. However, the AFWS baseline is almost a blank FCOS pipeline without whistles and bells. It can be noticed that the 1×-schedule, 3×-schedule, and 1×-schedule+MS AP_{50} of AFWS baseline comprehensively exceeds that of FCOS. The plug-and-play AFWS can certainly be incorporated into various advanced pipelines and hopefully improve performance. But even without considering this aspect, the fact that AFWS, as a WS method, is comparable to the fully supervised method is enough to demonstrate its effectiveness and value.

2) DIOR-R: As shown in Table II, the 3×-schedule AP_{50}, AP_{75}, and AP (at IoU = 0.50:0.05:0.95) of the AFWS baseline are 60.88%, 35.81%, and 36.69%, respectively, which are 0.46%, 0.33%, and 0.58% better than H2RBox-v2, respectively, reaching the SOTA among the WS methods. It is particularly noteworthy in this set of experiments that the gap between the WS method and the fully supervised method has narrowed dramatically, and it can even be said that the WS method is superior. In particular, the accuracy of H2RBox-v2 ranks second among the RBox-supervised methods (only lower than Oriented RepPoints), and that of AFWS baseline ranks fourth (lower than Oriented RepPoints, S^2A-Net, and CFA). This article believes that this is due to the presence of some unreasonable annotations in the DIOR-R dataset. For example, objects such as chimney, wind mill, and airport directly use the horizontal bbox as GT, which may be not consistent with the true orientation of the object. In other words, the results in Table II confirm that HBox-supervised methods are more robust than RBox-supervised methods when faced with incorrect rotating annotations, which is also logically

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TABLE II
TEST RESULTS ON THE DIOR-R DATASET

	Method	Sched.	Size	FPS	AP	AP_{50}	AP_{75}
RBox-supervised	RepPoints [35]	1×	800	31.8	32.41	55.24	30.71
	RetinaNet [24]	1×	800	33.5	32.13	54.58	31.21
	GWD [16]	1×	800	32.9	33.19	57.45	33.05
	KLD [36]	1×	800	33.3	33.24	56.69	32.64
	KFIoU [18]	1×	800	33.3	33.55	57.26	32.86
	SASM [37]	1×	800	31.5	33.34	58.57	33.25
	R^3Det [38]	1×	800	24.9	33.72	58.90	32.93
	CFA [39]	1×	800	31.8	34.25	58.43	33.38
	Oriented RepPoints [40]	1×	800	32.6	34.41	59.12	**34.03**
	S^2A-Net [41]	1×	800	28.6	**35.07**	**59.63**	33.61
	FCOS [34]	1×	800	**41.4**	31.81	55.16	29.65
	FCOS [34]	3×	800	**41.4**	33.25	55.91	32.11
HBox-supervised	BoxInst-RBox [7][1]	1×	800	3.5	31.73	57.40	28.10
	BoxLevelSet-RBox [8][1]*	1×	800	6.7	29.96	56.56	24.36
	SAM-ViT-B-RBox [9][1]	1×	800	2.3	32.37	57.61	30.52
	H2RBox (FCOS-based) [5]	1×	800	34.4	32.63	56.40	31.49
	H2RBox (FCOS-based) [5]	3×	800	34.4	35.02	57.82	34.22
	H2RBox-v2 (FCOS-based) [6]	1×	800	32.3	34.57	59.50	33.85
	H2RBox-v2 (FCOS-based) [6]	3×	800	32.3	36.11	60.42	35.48
	AFWS (FCOS-based)	1×	800	34.2	34.30	59.07	33.29
	AFWS (FCOS-based)	3×	800	34.2	**36.69**	**60.88**	**35.81**

[1] "-RBox" means post-processing the segmentation mask to obtain its minimum circumscribed rotation rectangle.
* An NVIDIA A100 GPU is used due to the significant memory requirement of the model.

TABLE III
AP_{50} PERFORMANCE ON THE DOTA-V1.5, DOTA-V2.0, AND FAIR1M DATASETS

	Method	Sched.	Size	DOTA-v1.5	DOTA-v2.0	FAIR1M
RBox-supervised	RepPoints [35]	1×	1024	60.48	47.90	37.80
	RetinaNet [24]	1×	1024	59.72	47.23	37.67
	GWD [16]	1×	1024	63.40	49.21	39.37
	KLD [36]	1×	1024	61.07	48.59	38.54
	KFIoU [18]	1×	1024	62.89	50.18	39.19
	SASM [37]	1×	1024	65.67	52.03	41.14
	R^3Det [38]	1×	1024	66.12	51.90	42.26
	CFA [39]	1×	1024	65.80	51.77	41.84
	Oriented RepPoints [40]	1×	1024	66.15	52.25	42.05
	S^2A-Net [41]	1×	1024	**66.76**	**52.70**	**42.52**
	FCOS [34]	1×	1024	62.38	49.88	39.11
	FCOS [34]	3×	1024	65.05	52.07	41.73
HBox-supervised	BoxInst-RBox [7][1]	1×	1024	46.70	31.83	24.58
	BoxLevelSet-RBox [8][1]*	1×	1024	49.08	36.35	27.97
	SAM-ViT-B-RBox [9][1]	1×	1024	55.46	47.15	37.84
	H2RBox (FCOS-based) [5]	1×	1024	62.18	50.07	37.35
	H2RBox (FCOS-based) [5]	3×	1024	64.24	51.92	40.16
	H2RBox-v2 (FCOS-based) [6]	1×	1024	64.49	50.65	42.40
	H2RBox-v2 (FCOS-based) [6]	3×	1024	66.10	52.44	44.11
	AFWS (FCOS-based)	1×	1024	65.92	51.73	41.80
	AFWS (FCOS-based)	3×	1024	**68.30**	**54.43**	**44.56**

[1] "-RBox" means post-processing the segmentation mask to obtain its minimum circumscribed rotation rectangle.
* An NVIDIA A100 GPU is used due to the significant memory requirement of the model.

self-consistent. In addition, another point to note is that the 1×-schedule accuracy of AFWS baseline is generally inferior to H2RBox-v2. This article believes that this is because H2RBox-v2 adds a symmetric view for SS training, which makes H2RBox-v2 more robust when facing incorrect annotations. However, the AFWS baseline has surpassed H2RBox-v2 on 3×-schedule accuracy, which confirms that the AFWS framework is indeed easier to tap the potential of the model, that is, the AFWS model is easier to train.

3) DOTA-v1.5/2.0 and FAIR1M: Table III records the AP_{50} performances of related works on the three datasets. Since DOTA-v1.5/2.0 adds a large number of small objects on the basis of DOTA-v1.0, the accuracy advantage of AFWS baseline compared to H2RBox-v2 is further demonstrated, with 1×-schedule improvements of 1.43% and 1.08%, and 3×-schedule improvements of 2.20% and 1.99% on DOTA-v1.5 and DOTA-v2.0, respectively. However, the AFWS baseline cannot outperform H2RBox-v2 on FAIR1M because FAIR1M contains a large number of strongly symmetric objects (such as airplanes and ships), which makes H2RBox-v2's symmetry learning take advantage. In addition, compared with the RBox-supervised methods, the rankings of H2RBox and H2RBox-v2 on DOTA-v1.5 and DOTA-v2.0 have declined compared to DOTA-v1.0. In particular, H2RBox ranks last among the RBox-supervised methods on FAIR1M. In contrast, the AFWS baseline maintains its ranking and narrows

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the accuracy gap with the RBox-supervised methods. This confirms that the complexity of the H2RBox series pipeline is a disadvantage when faced with difficult tasks such as small object detection (DOTA-v1.5/2.0) and fine-grained classification (FAIR1M). While the advantages of AFWS in simplifying the training process and optimizing the loss function are highlighted in this dilemma, which is also reflected in the fact that AFWS's 3×-schedule AP_{50} surpasses that of H2RBox-v2.

4) Analysis on the Superiority of AFWS: This article believes that the accuracy of AFWS can surpass H2RBox/H2RBox-v2 mainly due to the superiority of lite-CCH representation. *First*, it enables the WS branch of AFWS to focus only on the training of horizontal parameters, making it easier for the model to reach a good and stable state. Specifically, as mentioned in the introduction, since the angle and horizontal parameter are strongly coupled in the angle-based RBox representation, H2RBox/H2RBox-v2 has to train a full set of RBox parameters (x, y, w, h, θ) in the WS branch and therefore faces the limitations of angle-based regression. In addition, H2RBox-v2 encounters the interplay between three views. All of them have greatly increased the difficulty of training, posing strong challenges to improving the accuracy and stability of the model. In contrast, the WS branch of AFWS is completely an ordinary horizontal object detection pipeline. This is due to the fact that the lite-CCH representation can formally decouple the horizontal and rotating parameters. The two can each be independently used as an aspect to describe the rotating object, even if they are separated, it is meaningful. At the same time, the rotating object detection accuracy based on the lite-CCH representation is based on the accuracy of its horizontal object detection, and it is well known that the horizontal object detection model is easy to train to achieve excellent performance. Therefore, this article believes that the WS branch of AFWS only focuses on the horizontal parameter training which is the main reason for its accuracy advantage. It is not difficult to understand logically. Generally speaking, when faced with a challenging problem, having too many learning objectives will make it more difficult to cover everything, distracting attention and losing sight of the other. It is usually easier to achieve success when learning objectives are few and clear. *Second*, as discussed previously, H2RBox and H2RBox-v2 perform a two-step approximation of the RBox IoU loss, and their approximation strategies are not closely related to the logic of the RBox IoU loss, resulting in a gap between the training objectives and the testing metrics. In contrast, AFWS only uses circular IoU in the SS branch as an approximation for the RBox IoU. Please note that although L_{ss} contains three parts, the HBox IoU loss only plays a weak role because it has been trained to a rather small value in the WS branch. In fact, subsequent ablation studies confirmed that removing the HBox IoU from L_{ss} would not have an important impact. Moreover, although L_{reg}^{or} is in the form of regression, it is essentially a discrete classification problem, which is specifically solved by the classification subnet. Therefore, only the one-step approximation of circular IoU actually affects the RBox IoU. It complements the lite-CCH representation and provides a more direct and intuitive alternative to the RBox IoU. It is logically smooth and subsequent ablation studies also confirm its superiority.

TABLE IV
TEST RESULTS ON DOTA-V1.0 FOR WHETHER THE SS BRANCH CONTAINS TRAINING OF CATEGORY OR HORIZONTAL PARAMETERS

L_{cls}	L_{iou}^{hb}	**AP**	**AP_{50}**	**AP_{75}**
✓	✓	40.50	71.83	39.21
✓		38.09	69.79	36.80
	✓	**41.02**	**72.55**	**39.86**
		39.32	71.17	38.04

D. Ablation Studies

All ablation studies in this section are conducted on DOTA-V1.0. In each ablation study, other details except the research content are the same as in AFWS baseline.

1) Impact of Whether the Object Category and Horizontal Regression Parameters Participate in Training in the SS Branch: What must be involved in L_{ss} is L_{iou}^{cr} and L_{reg}^{or}. As for L_{cls} and L_{iou}^{hb}, they are included in L_{ws}. The impact on accuracy caused by adding L_{cls} and L_{iou}^{hb} to L_{ss} is shown in Table IV. The third row in the table is the test results of AFWS baseline, that is, L_{ss} contains L_{iou}^{hb} but does not involve L_{cls}. From this ablation study, it can be concluded as follows.

1) The participation of the horizontal parameter in training in the SS branch will play an icing on the cake role and improve the accuracy. However, even if L_{iou}^{hb} is not involved in L_{ss}, it will not have a disruptive impact. As shown in the fourth row of results, its accuracy has a loss but not much. Putting it in Table I is still competitive. The above results also confirm that the WS branch plays an absolutely dominant role in horizontal parameter training.
2) The category parameter participating in training in the SS branch will not improve accuracy but have the opposite effect. This result also confirms the analytical discussion in Section III-B, that is, the core task of the SS branch is the training of rotating parameter, especially the circle parameter, which is quite challenging. Having additional objectives that distract from the core task can indeed be counterproductive.

2) Role of Circular IoU Loss: This ablation experiment studies the impact of using the $L1$ loss, BCE loss, and circular IoU loss to train the cr on accuracy. The results are presented in Table V. Results show that, for AFWS, the circular IoU loss has obvious superiority, and its accuracy is significantly better than the $L1$ loss and BCE loss. Based on the previous discussion, $L1$ loss, BCE loss, and circular IoU loss are all one-step approximations to the RBox IoU loss in form, but the logical connection between circular IoU loss and RBox IoU loss is obviously closer than the other two. In other words, the $L1$ loss and BCE loss are just an evaluation of the numerical accuracy of the cr prediction, while the RBox IoU loss has geometric meaning. There is still a conceptual gap between the two, and the correlation is not direct enough. However, the geometric meaning of circular IoU loss is rather intuitive, and its impact on the RBox IoU loss is more direct, which is consistent with the analysis in the previous section.

3) Setting of the Hyperparameters in the Training Objective: This ablation experiment only studies the impact of the two weights of λ and γ_1 on accuracy. This is because the WS branch is actually an FCOS pipeline, so the settings

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TABLE V
TEST RESULTS ON DOTA-V1.0 USING DIFFERENT LOSSES FOR THE CIRCLE PARAMETER

Loss for *cr*	**AP**	**AP$_{50}$**	**AP$_{75}$**
L1	33.52	62.87	32.10
BCE	36.48	66.91	35.35
circular IoU	**41.02**	**72.55**	**39.86**

TABLE VI
TEST RESULTS ON DOTA-V1.0 BY DIFFERENT HYPERPARAMETER SETTINGS

λ	γ_1	**AP**	**AP$_{50}$**	**AP$_{75}$**
1	2	28.90	58.70	26.48
	1	23.73	51.54	18.04
0.5	2	35.14	64.51	33.72
	1	31.29	60.45	29.46
0.25	2	**41.02**	**72.55**	**39.86**
	1	37.86	66.98	36.39
0.1	2	29.30	58.63	25.78
	1	24.52	52.13	20.91

of β_1 and β_2 can follow FCOS; and γ_2 plays a secondary role in the SS branch. The corresponding test results are given in Table VI. First, in terms of the dominance of WS branch and SS branch, the test results of different λ settings show that, when the WS branch occupies a certain degree of dominance, it will promote the improvement of accuracy. This is because the overall detection effect of AFWS is based on the effect of horizontal object detection. The test results confirm the previous analysis and are consistent with the principle of the lite-CCH representation. Second, the ablation study on γ_1 confirms that AFWS can achieve better performance when the circular IoU loss dominates L_{ss}. That is, the SS branch should indeed focus mainly on the training of cr, which also confirms the rationality of the overall design concept of AFWS.

V. CONCLUSION

This article deeply analyzes the limitations of HBox-based WS rotating object detection caused by angle regression, and proposes a WS rotating object detection framework based on an improved angle-free rotating object representation. The proposed AFWS decouples the training of horizontal and rotating regression parameters, simplifying the dominant WS branch into a pure common horizontal object detection pipeline, thereby effectively reducing the difficulty of model training. In addition, the introduced SS loss function establishes a closer geometric and logical connection with the RBox IoU loss, bridging the gap between the training objective and the testing metrics to achieve improved accuracy. However, the SS loss of AFWS still involves angle and absolute value terms and is therefore not completely differentiable, and its SS branch essentially implies horizontal parameter training and therefore does not achieve complete decoupling. These two points are expected to be further improved in subsequent work.

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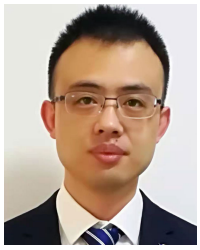
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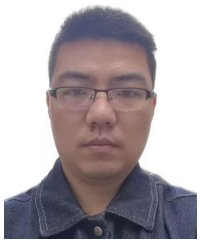
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